



Retail Sales Analysis Using MySQL

A Data Analytics Portfolio Project focused on transforming retail sales data into actionable business insights using MySQL.

Data Analytics

SQL



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Project Objective



Analyze retail sales data to uncover customer behavior, identify sales trends, and generate actionable business insights using SQL.

Dataset Overview

- **Dataset:** Retail
- **Records:** 2000
- **Columns:** 11
- **Categories:** Clothing, Beauty, Elec.
- **DB:** MySQL

Skills Demonstrated

- ✓ MySQL & SQL Querying
- ✓ Data Cleaning & Exploration
- ✓ Aggregate & Window Functions
- ✓ Sales Trend Analysis
- ✓ Business Insight Generation

Purpose: This dataset contains customer purchase transactions used to analyze sales performance, customer behavior, and business trends.



SQL Analysis Breakdown



Basic Analysis

- Sales transactions on 5th November 2022
- Clothing transactions with quantity > 4 in November 2022
- Average age of Beauty category customers
- High-value transactions (>1000)

Intermediate Analysis

- Total sales revenue and orders by category
- Gender-wise transactions by category
- Number of unique customers by category
- Top 5 customers *by purchase value*

Advanced Analysis

- Average monthly sales & best-selling month
- Shift-wise order analysis (Morning, Afternoon, Evening)



Write a SQL query to find the number of unique customers who purchased items from each category

```
SELECT
    category,
    COUNT(DISTINCT customer_id) AS unique_customers
FROM retail_sales
GROUP BY category;
```

Result Grid   Filter Rows:		
	category	unique_customers
▶	Beauty	141
	Clothing	149
	Electronics	144



Write a SQL query to calculate the total sales (total_sale) for each category.

```
SELECT
    SUM(total_sale) AS net_sales, category
FROM
    retail_sales
GROUP BY category
ORDER BY category DESC;
```

Result Grid   Filter R		
	net_sales	category
▶	311445.00	Electronics
	309995.00	Clothing
	286790.00	Beauty



Write a SQL query to create each shift and number of orders (Example Morning <12, Afternoon between 12 & 17, Evening >17)

```
SELECT
  CASE
    WHEN HOUR(sale_time) < 12 THEN 'Morning'
    WHEN HOUR(sale_time) BETWEEN 12 AND 17 THEN 'Afternoon'
    ELSE 'Evening'
  END AS shift,
  COUNT(*) AS number_of_orders
FROM retail_sales
GROUP BY shift;
```

Result Grid			Filter Rows:
	shift	number_of_orders	
▶	Evening	1062	
	Morning	548	
	Afternoon	377	



Write a SQL query to calculate the average sale for each month. Find out best selling month in each year

```
SELECT
    AVG(total_sale) AS avg_sales
    YEAR(sale_date) AS year,
    MONTH(sale_date) AS month
FROM retail_sales
GROUP BY 2, 3;
```

Result Grid				Filter Rows:
	avg_sales	year	month	
▶	460.769231	2022	12	
	481.395349	2022	6	
	495.964912	2023	8	
	462.739726	2023	9	
	453.452381	2023	11	
	427.678571	2023	7	
	467.137931	2022	10	
	366.136364	2022	2	
	535.531915	2023	2	
	480.000000	2023	5	

Business Insights



- ✓  Electronics generated the highest revenue, indicating strong customer demand.
- ✓  November was the best-performing month, representing a peak sales period.
- ✓  Evening hours recorded the highest order volume, suggesting increased customer activity.
- ✓  Customer ID 102 contributed the highest total purchase value.
- ✓  Clothing accounted for the highest transaction volume among all categories.



Project Summary

Built end-to-end SQL project: from database creation and cleaning to analytics and business recommendations.

- ✓ Analyzed retail sales data using MySQL.
- ✓ Identified sales trends and customer behavior.
- ✓ Evaluated category performance and top customers.
- ✓ Applied SQL functions to generate business insights.
- ✓ Enabled data-driven decision-making through analytical reporting.

GitHub: <https://github.com/AkshatNigam06>

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Thank you — happy to walk through queries, schemas, and results in a follow-up demo